We can try to generalize the concept of intrinsic/extrinsic

under different modalities

Modality	Input	Intrinsic	Extrinsics
2D	Paired Images with the same scene	Albedo, local geometric	Lighting
3D	Multi-view Images of the same Scene	3D geometric, Albedo	Camera, Lighting
4D	Frames of video sequences	3D geometric, Albedo	Camera, Lighting, Object Motion/ Deformation

Goal:

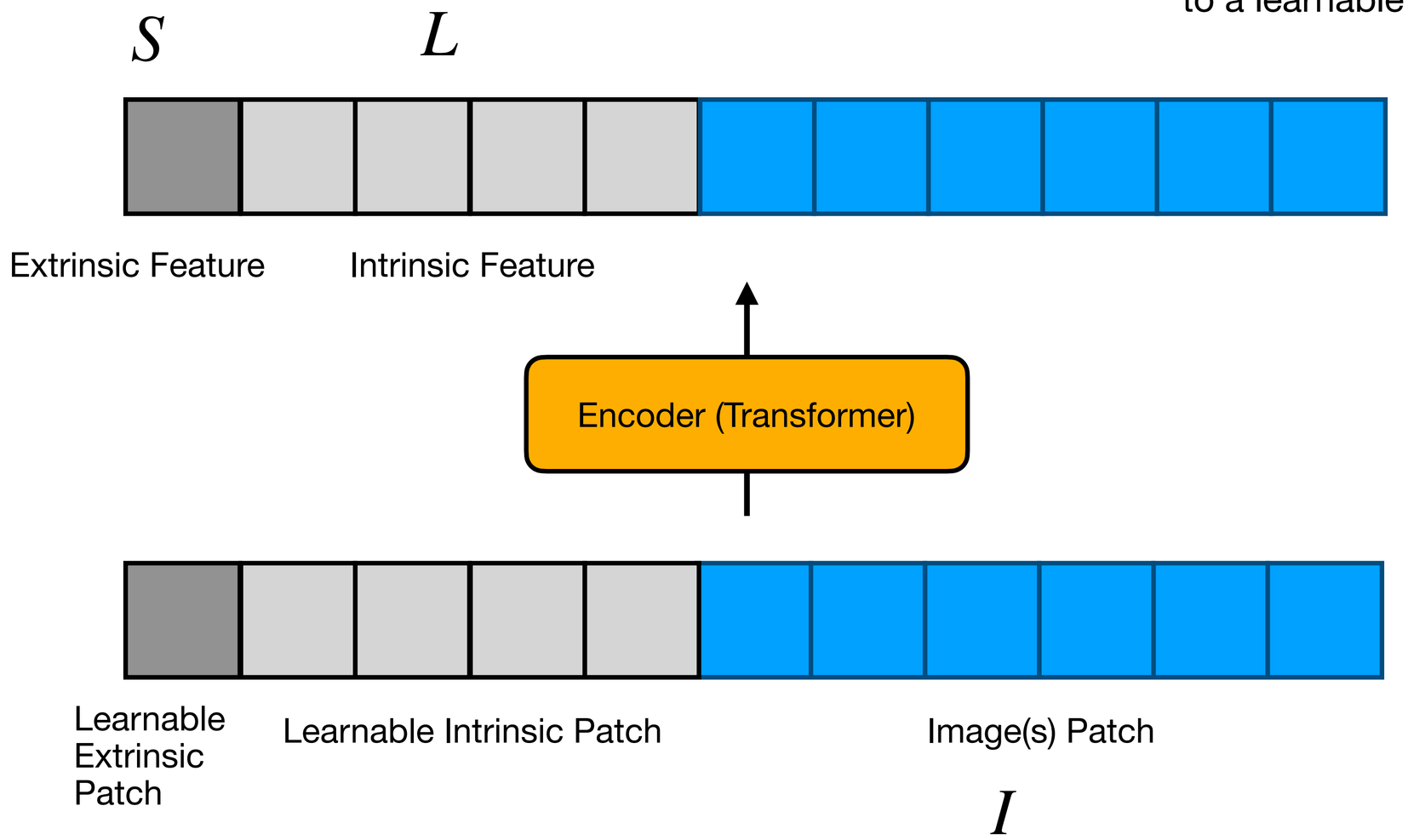
Generalize the architecture of our Neurips submission. to learn latent intrinsic and extrinsic for different modalities to solve the the issue of requiring paired data

In the new framework, we still have an encoder that maps input to intrinsics and extrinsic as latent representations.

The loss is the same — reconstruction loss and intrinsic invariant loss

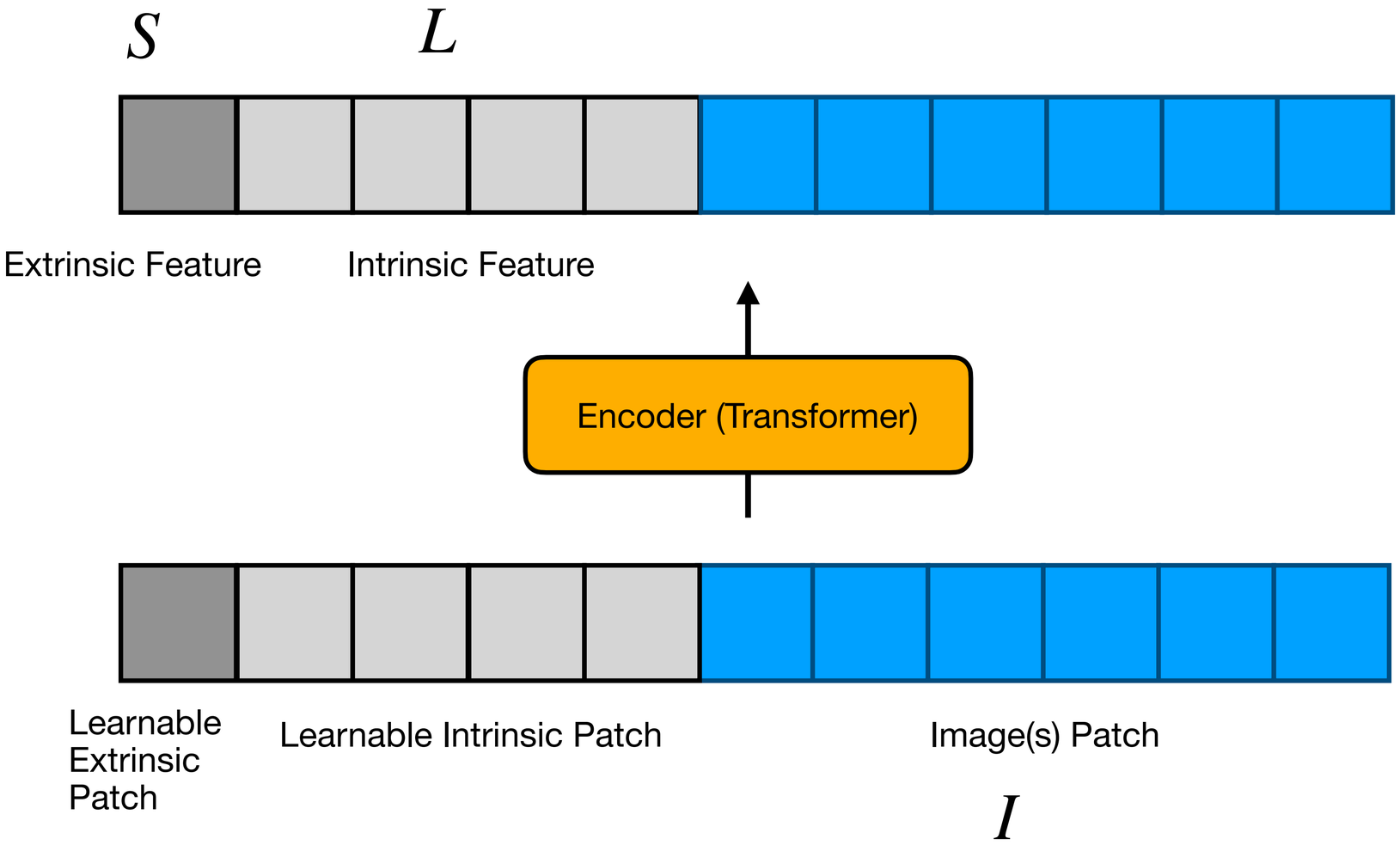
Network Architecture: Encoder

Encoder is now a transformer that maps input image to a learnable tokens for intrinsic and extrinsic.



$$E(I) = L, S$$

Case1: 2D, the same as our Neurips submission.



I_1, I_2 are paired images from the same scene

$$E(I_1) = L_1, S_1$$

$$E(I_2) = L_2, S_2$$

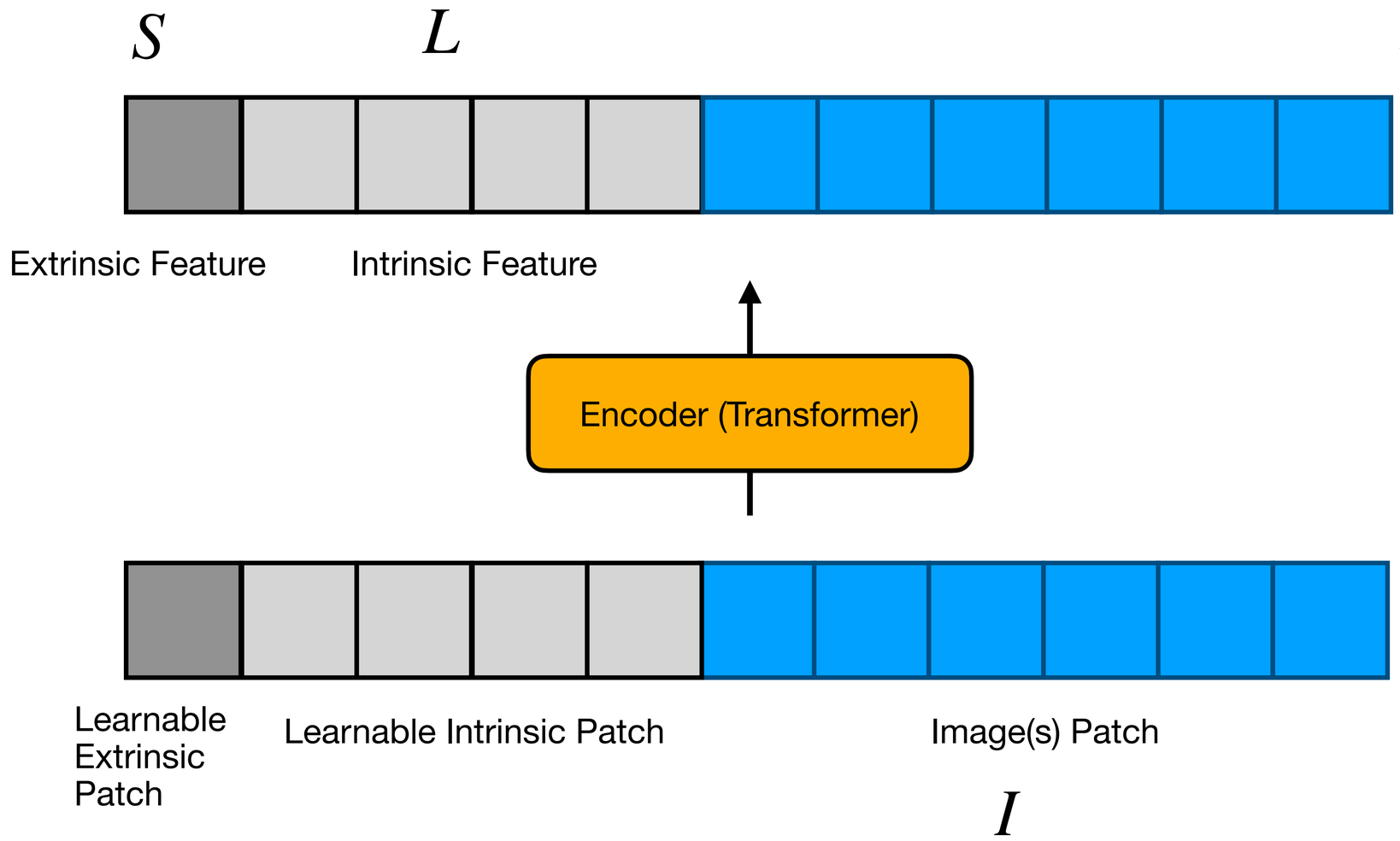
$$D(L_1, S_2) = \hat{I}_2$$

$$\mathscr{L} = \|L_1 - L_2\| + \|I_2 - \hat{I}_2\|$$

S_1, S_2 **encode Extrinsic Light**

$L_1 = L_2$ **encodes albedo + local geometric**

Case2: 3D



$$E(I) = L, S$$

I_1, I_2 are multi-view image(s)* from the same scen

$$E(I_1) = L_1, S_1$$

$$E(I_2) = L_2, S_2$$

$$D(L_1, S_2) = \hat{I}_2 \text{ **}$$

$$\mathscr{L} = \|L_1 - L_2\| + \|I_2 - \hat{I}_2\|$$

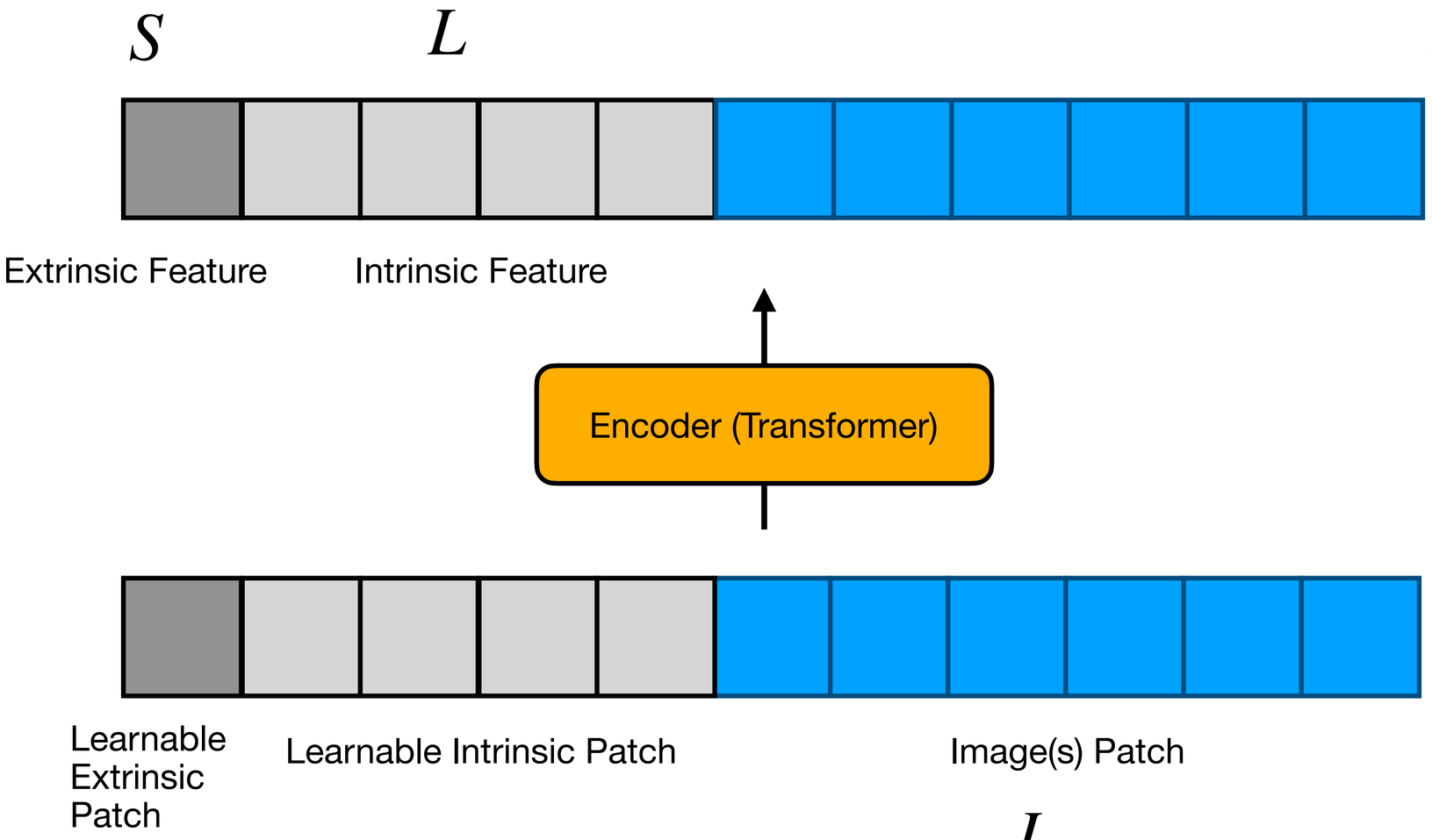
S_1, S_2 **Encode Extrinsic Light and Camera**

$L_1 = L_2$ **encodes albedo + 3d geometric**

*I_1, I_2 could be single or multiple images to deal with ambiguity

** Reconstruction loss can be replaced with diffusion loss to deal with ambiguity. $D(L_1, S_2, I_2 + \epsilon) = \hat{I}_2$

Case2: 4D



$$E(I) = L, S$$

I_1, I_2 are frame(s) from the video clip

$$E(I_1) = L_1, S_1$$

$$E(I_2) = L_2, S_2$$

$$D(L_1, S_2) = \hat{I}_2$$

$$\mathscr{L} = \|L_1 - L_2\| + \|I_2 - \hat{I}_2\|$$

S_1, S_2 **Encode Extrinsic Light and Camera and object motion/deformation**

$L_1 = L_2$ **encodes albedo + 3d geometric**